



LLM Safety Harness

Responsible AI
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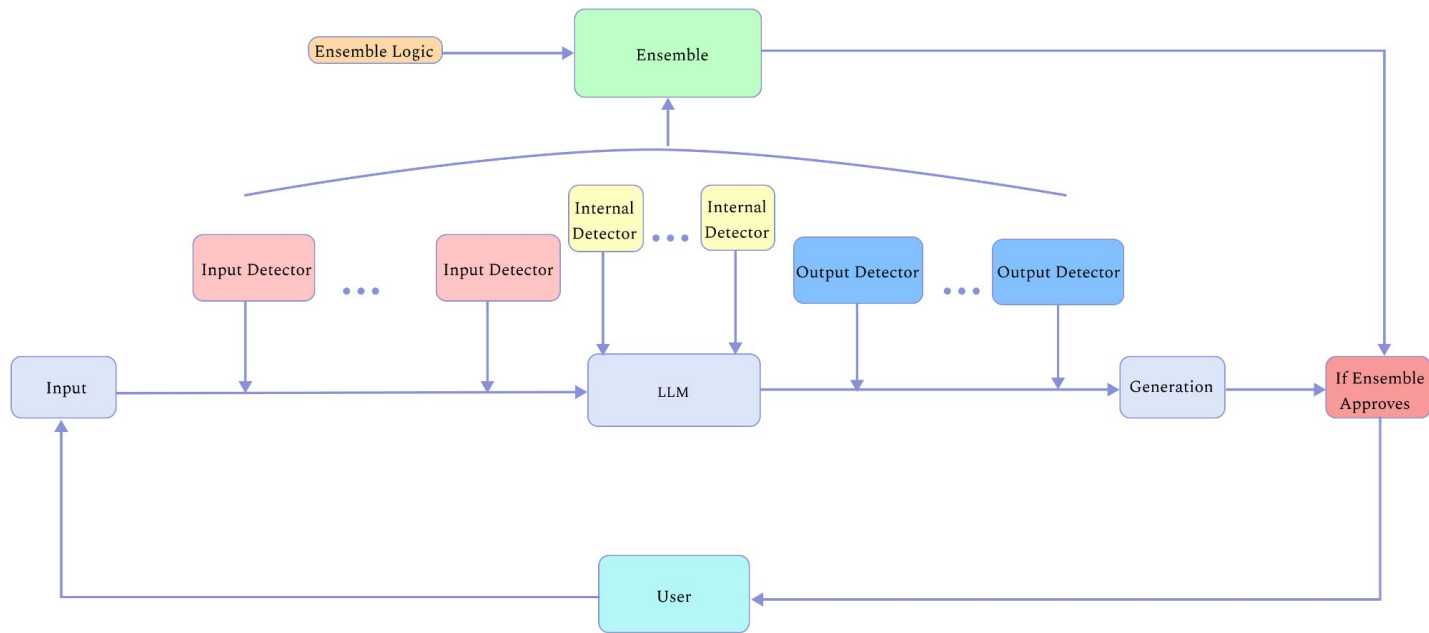
Problem LLMs are not safe

- Jailbreak: Breaking its safeguards
- SOTA models not safe - even more dangerous
- Mitigation strategies → High False Positive, Overrefusal

Overview

- Comparison of BERT models for Jailbreak detection
- Comparing BERT classifiers to linear probes
- Strengths and trade-offs of AND ensemble classifier

Architecture



- Protective scaffold around LLM
- Allows for multiple detectors

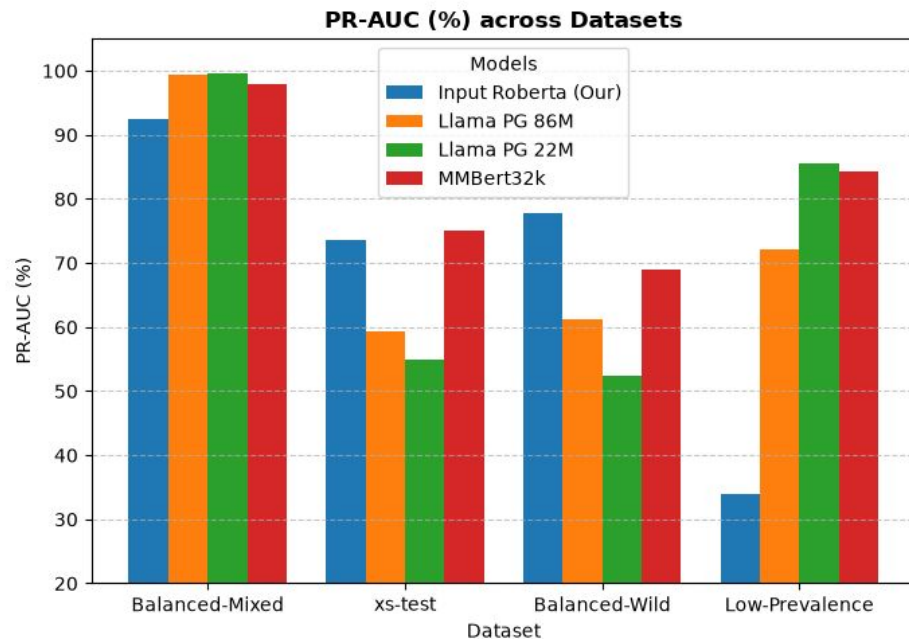
Input/Output Classifier

- BERT models to capture sentiment
- Binary Classifiers → 1 harmful and 0 benign
- Challenges:
 - Very high refusal rate
- Solution: Label smoothing, data augmentation

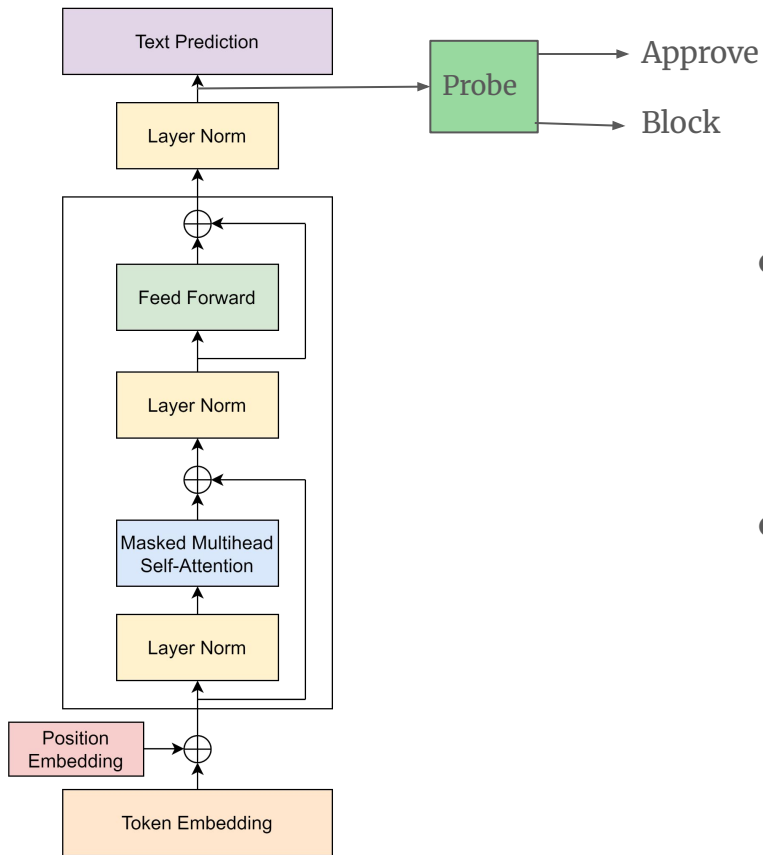
Input BERT comparison

- 1 realistic split: 99% benign, 1% adversarial
- 3 balanced 50/50 sets with different challenges
- BERT classifier threshold dependent
- Precision Recall Curve → threshold-agnostic

Input BERT PR-AUC



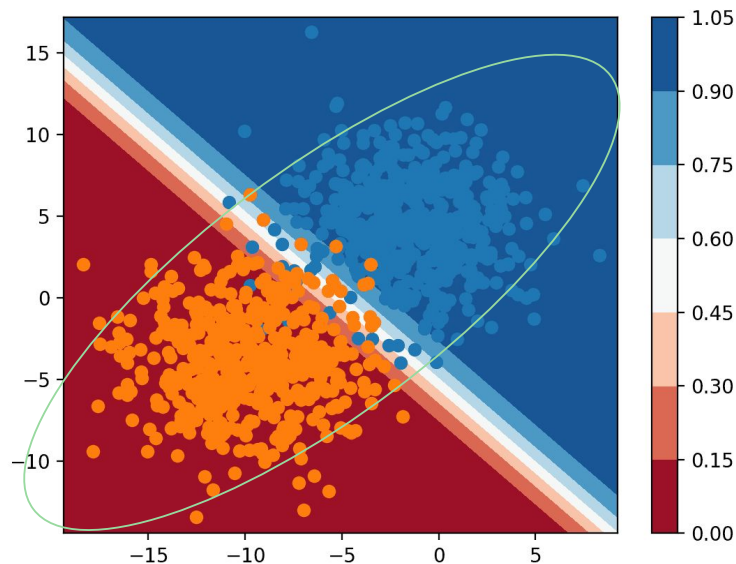
Internal Classifier



- Instead of text, we classify **activations of LLM**
- Key advantage: Allows usage of tiny neural networks, no BERT needed

Internal Classifier

- Choice: Linear Probes
- Why? Light weight, Linear Representation Hypothesis (LRH)
- LRH: Features are approximately linear in the latent space
- Classify **Input** activations only

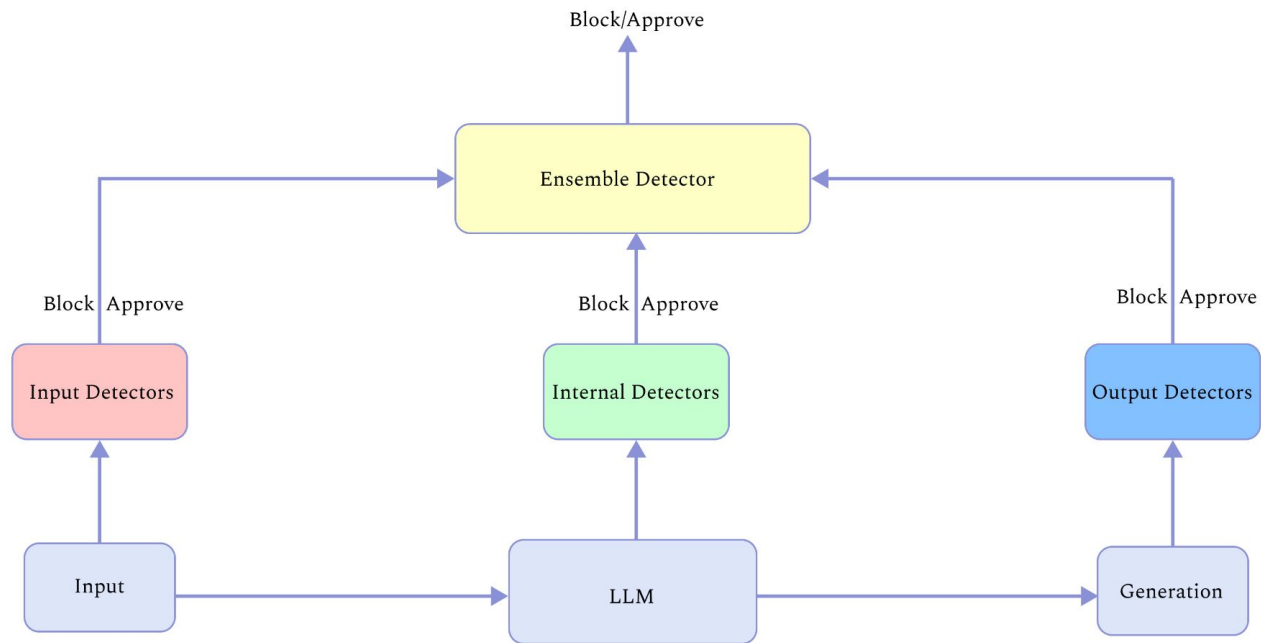


<https://machinelearningmastery.com/plot-a-decision-surface-for-machine-learning/>

Latency Comparison

Latency	Median (ms)
roBERTa Input	13.35
Linear Probe	0.0523
roBERTa Output	64.81
Ensemble System	78.2123

Ensemble



- And logic: $\text{Input} \wedge \text{Internal} \wedge \text{Output}$

Ensemble

Input Label	Output Label	Combined label
0	0	0 (Pass)
0	1	0 (Pass)
1	0	0 (Pass)
1	1	1 (Block)

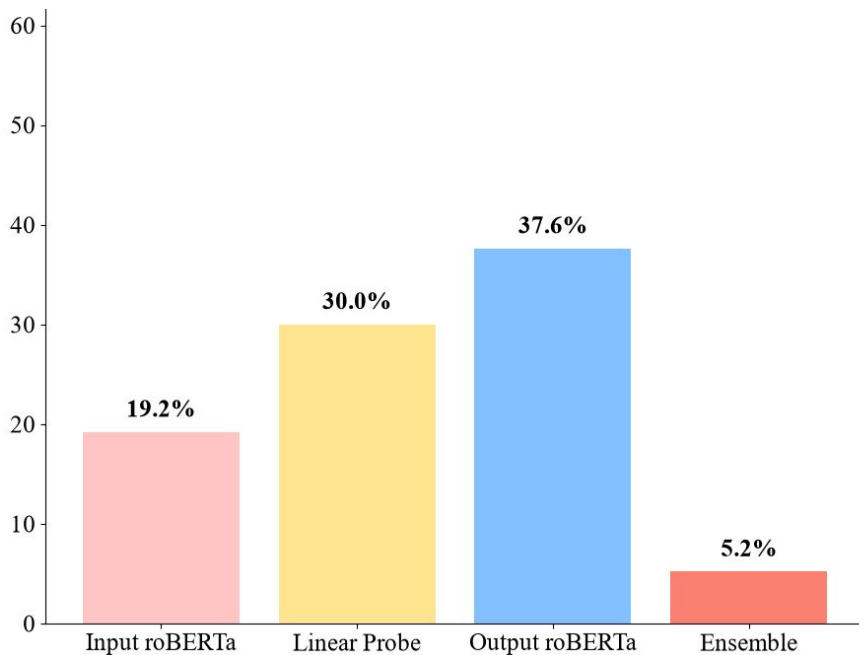
- We only care about harmful inputs that result in harmful outputs
- Send output to user when:
 - harmful hallucinations (0,1) → Unlikely
 - LLM output harmless (1,0)

Experiment Setup

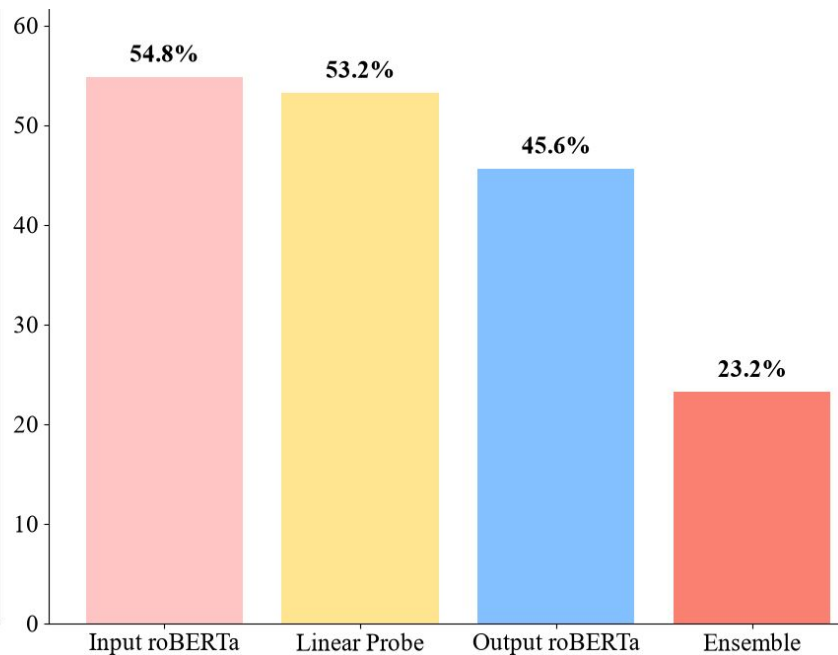
- Four scenarios:
 - Input detector
 - Internal detector
 - Output detector
 - Ensemble
- **Overrefusal Rate:** only benign → measure falsely blocked
- **Attack success rate:** only harmful → measure amount passed

Overrefusal experiment

Alpaca Cleaned

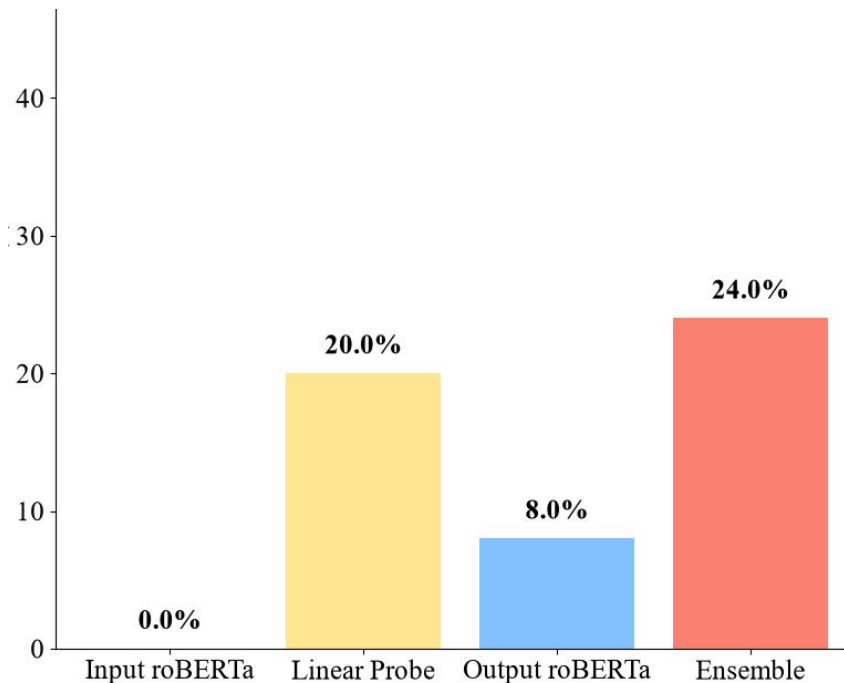


XS-Test (Benign)

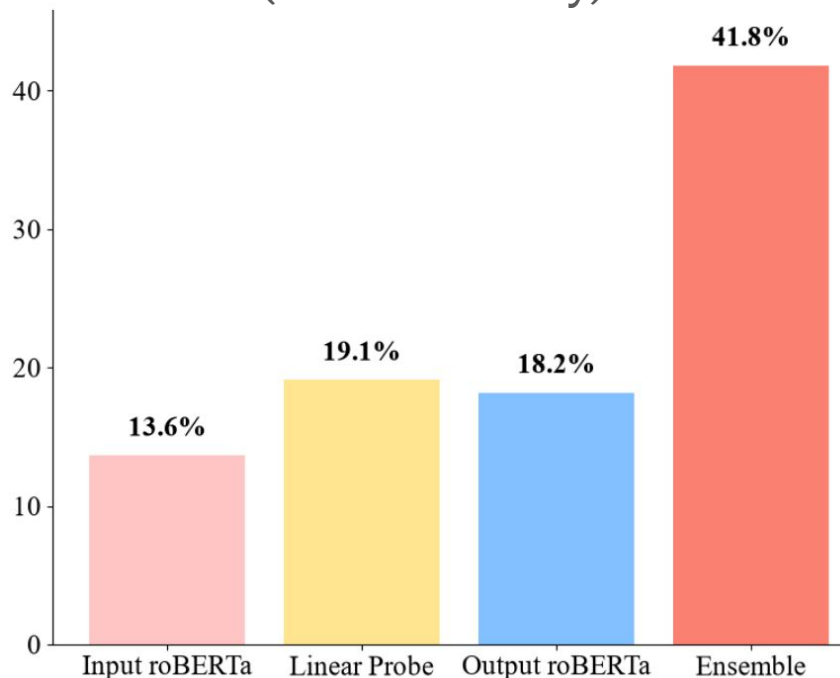


Attack Success Rate experiment

AdvBench



WildJailbreak Adversarial
(Harmful Only)



Conclusion

- Bert Models and LP both have their advantages and disadvantages
 - Bert accuracy
 - LP speed
- All classifiers struggle with Overrefusal
- AND strategy
 - helps reduce the ORR considerably, increases ASR
 - recommended for low ASR, high ORR models

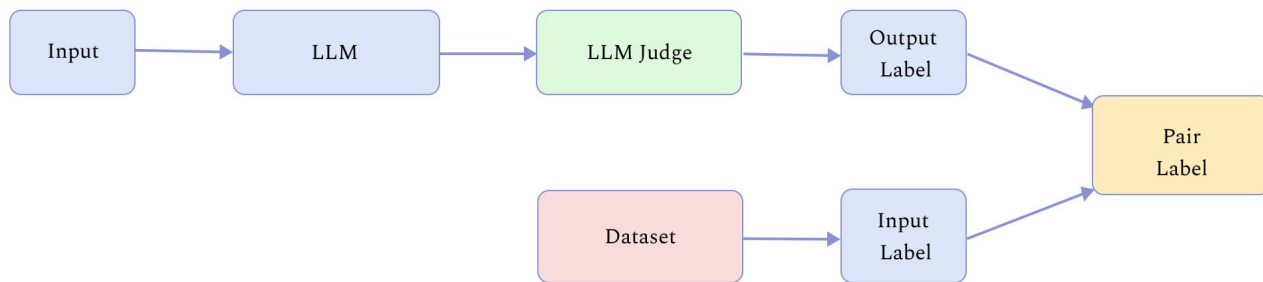


Questions

ROC-AUC and PR-AUC

- ROC-AUC
 - TPR vs FPR
 - If Harmful = Label 1 | Safe = Label 0
 - Attack Detection vs Over Refusal Rate
 - How much do I have to overrefuse to catch more attacks?
- PR-AUC
 - Precision vs Recall
 - Refusal Reliability vs Attack Detection
 - How many blocks were innocent users?

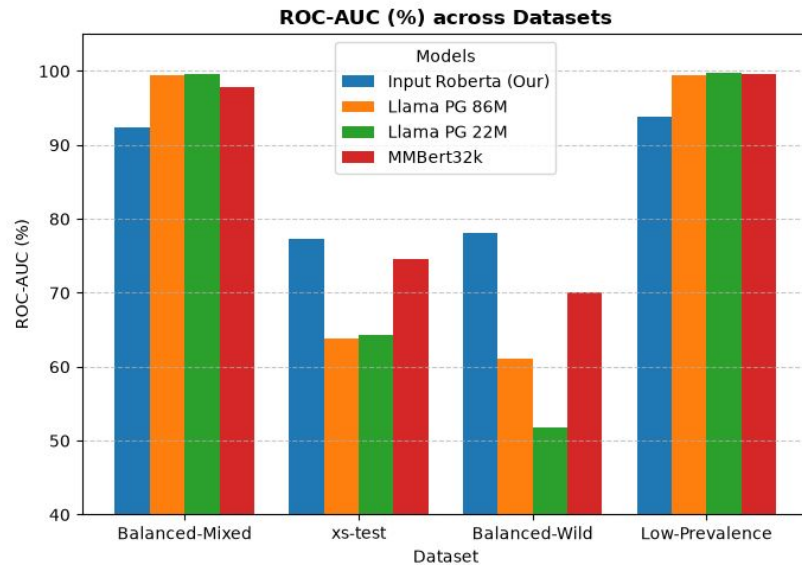
Ensemble: Gold Label



Input Label	Output Label	Combined label
0	0	0 (Pass)
0	1	0 (Pass)
1	0	0 (Pass)
1	1	1 (Block)

- Pass Dataset through the LLM,
- Label output with LLM-as-a-judge

Input BERT Graphs



Evaluation Datasets

Alpaca-Cleaned: A dataset containing instruction prompts generated by GPT-3.5; they are generally short in length and cover simple tasks.

XS-test: Contains adversarial benign prompts specifically designed to test the overrefusal metric.

AdvBench: This is our baseline measure, it contains simple harmful prompts that attempt the model to engage in harmful behaviour.

Wildjailbreak Adversarial: The adversarial subset of the wildjailbreak dataset, deploys difficult adversarial strategies to "hide" the jailbreaking prompt.